

Problem Set 3

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Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

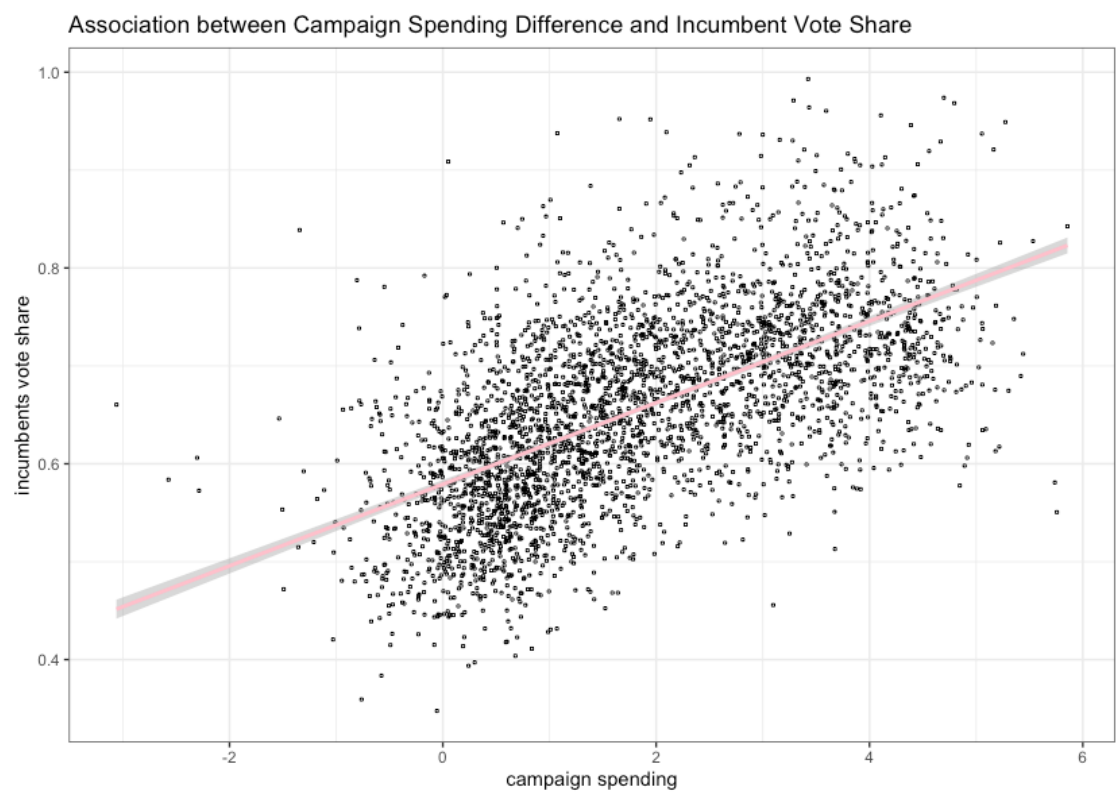
We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
head(inc.sub)
str(inc.sub)
regression1 <- lm(voteshare ~ difflog, data = inc.sub)
summary(regression1)
```

2. Make a scatterplot of the two variables and add the regression line.

```
ggplot(data = inc.sub, aes(x = difflog,  
y = voteshare)) +  
geom_point(size = 0.5, shape = 22) +  
geom_smooth(method = lm, color = "pink") +  
labs(title = "Association between Campaign  
Spending Difference and Incumbent Vote Share",  
x = "campaign spending",  
y = "incumbents vote share") +  
theme_bw()
```



3. Save the residuals of the model in a separate object.

```
residuals1 <- regression1$residuals  
residuals1
```

4. Write the prediction equation.

```
summary(regression1)
```

This summary gives us the following coefficients:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.579031	0.002251	257.19	<2e-16 ***
difflog	0.041666	0.000968	43.04	<2e-16 ***

We can now write the prediction equation $\gamma = \alpha + \beta * x$.

Incumbent vote share = intercept + slope * difflog

Incumbent vote share = 0.579031 + 0.041666 * difflog

On average, for every one-unit increase of the explanatory variable "difflog" which is campaign spending, the output variable "Incumbent vote share" increases by 0.041666.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

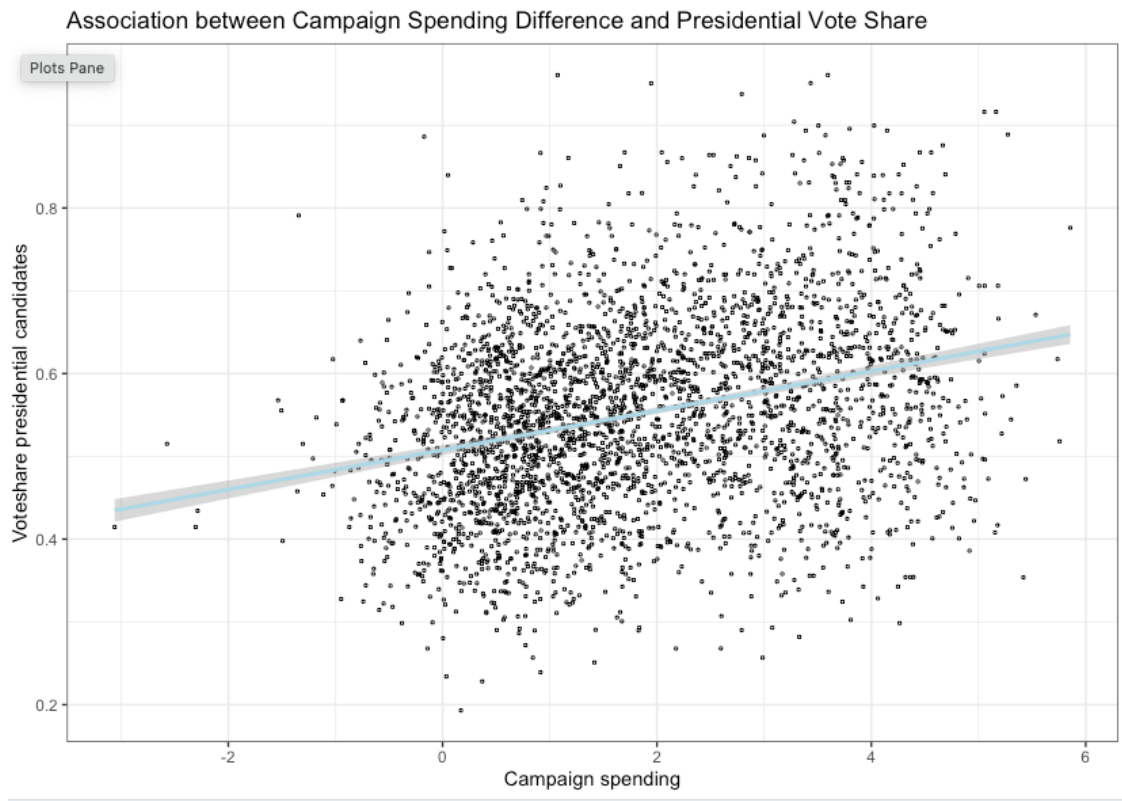
1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

How is the difference in spending related to the presidential candidates vote share?

```
regression2 <- lm(presvote ~ difflog, data = inc.sub)
summary(regression2)
```

2. Make a scatterplot of the two variables and add the regression line.

```
ggplot(data = inc.sub, aes(x = difflog, y = presvote)) +
  geom_point(size = 0.5, shape = 22) +
  geom_smooth(method = lm, color = "lightblue") +
  labs(title = "Association between Campaign Spending Difference
and Presidential Vote Share",
x = "Campaign spending",
y = "Voteshare presidential candidates") +
  theme_bw()
```



3. Save the residuals of the model in a separate object.

```
residuals2 <- regression2$residuals
residuals2
```

4. Write the prediction equation.

To find the prediction equation, we need to find the relevant coefficients:

```
summary(regression2)
```

Coefficients:

```
Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) 0.507583 0.003161 160.60 <2e-16 ***
```

```
difflog      0.023837 0.001359 17.54 <2e-16 ***
```

We can now write the prediction equation $\gamma = \alpha + \beta \cdot x$.

presidential candidate vote share = intercept + slope * difflog presidential candidate

vote share = 0.507583 + 0.023837 * difflog

In other words, on average, for every one-unit increase of the explanatory variable "difflog", which is campaign spending, the output variable "vote-share presidential candidate" increases by 0.023837.

Question 3

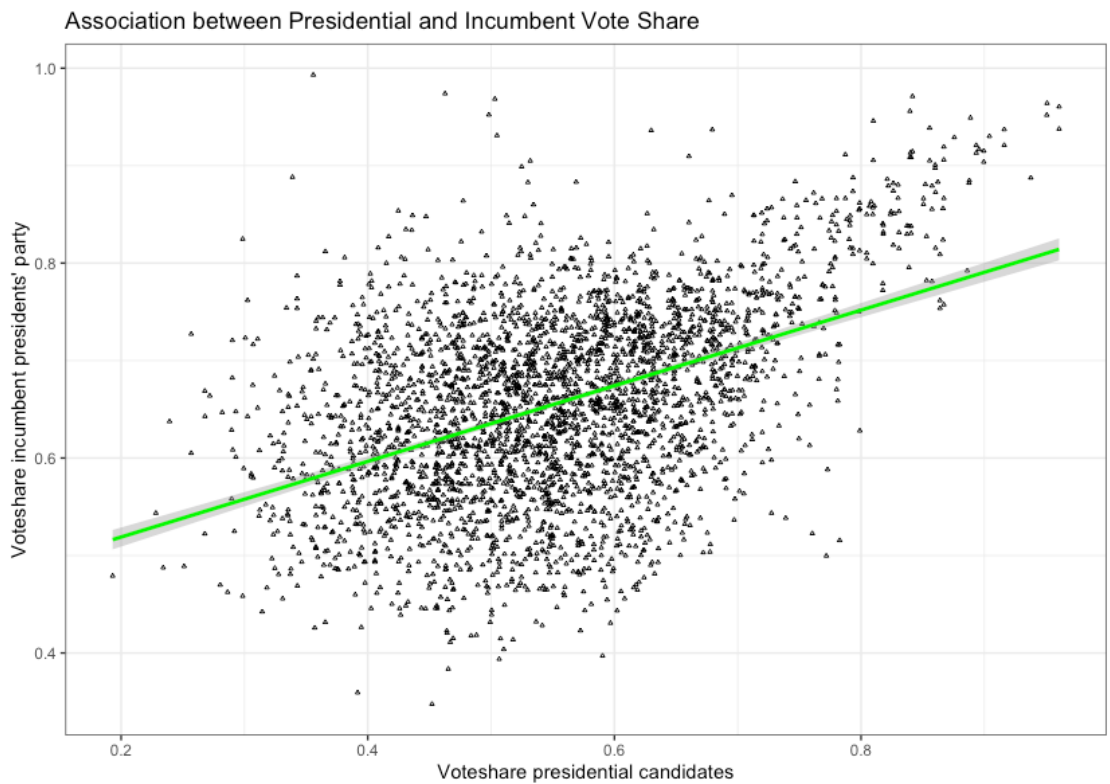
We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
regression3 <- lm(voteshare ~ presvote, data = inc.sub)
summary(regression3)
```

2. Make a scatterplot of the two variables and add the regression line.

```
ggplot(data = inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point(size = 0.5, shape = 2) +
  geom_smooth(method = lm, color = "green") +
  labs(title = "Association between Presidential and Incumbent Vote Share ",
       x = "Voteshare presidential candidates",
       y = "Voteshare incumbent presidents' party") +
  theme_bw()
```



3. Write the prediction equation.

To make the prediction equation, we need to find the relevant coefficients.

regression3 We get the following coefficients:

Coefficients:

Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept) 0.441330 0.007599 58.08 <2e-16 ***

presvote 0.388018 0.013493 28.76 <2e-16 ***

We can now write the prediction equation $\gamma = \alpha + \beta * x$.

Votes share incumbent presidents' party = Intercept + slope * votes share presidential candidate

Votes share incumbent presidents' party = $0.4413 + 0.3880 * \text{Votes share presidential candidate}$

On average, for every one-unit increase of the explanatory variable "presvote" which is the incumbents electoral success, the output variable "votes share", which is the vote share of the presidential candidate of the incumbent's party, increases by 0.3880.

Question 4

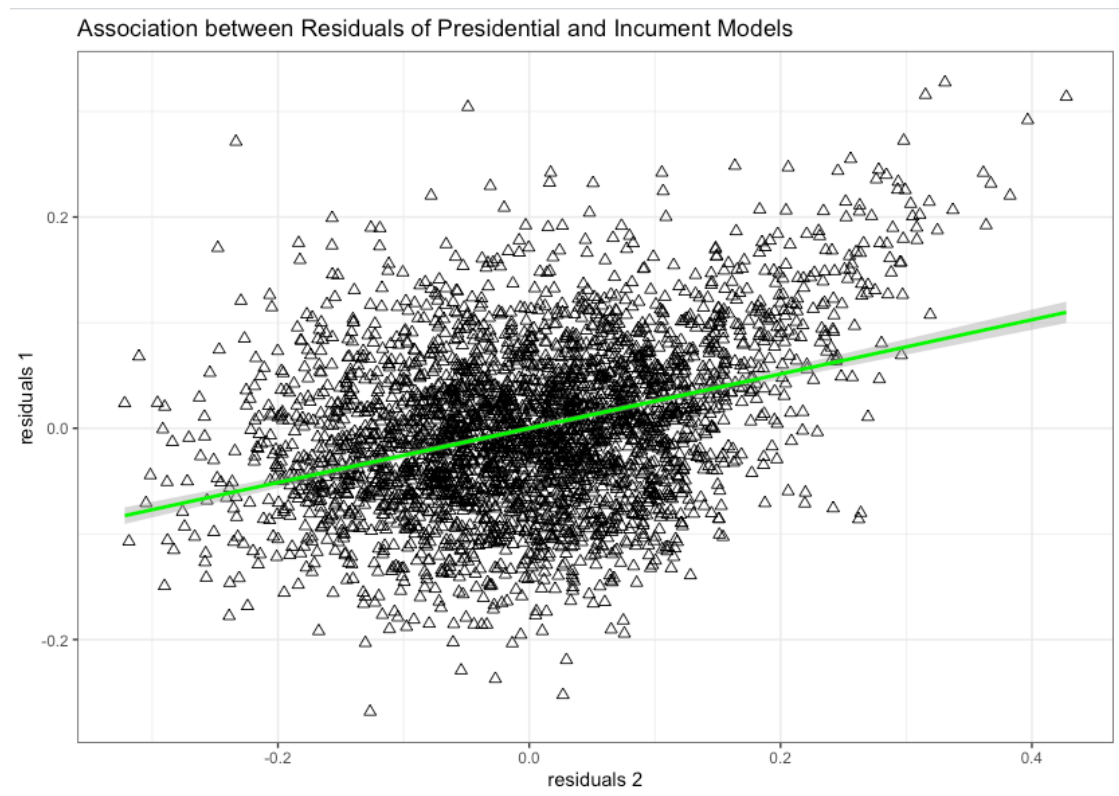
The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
dataframe <- data.frame(residuals_1 = residuals1, residuals_2 = residuals2)
regression4 <- lm(residuals_1 ~ residuals_2, data = dataframe)
```

2. Make a scatterplot of the two residuals and add the regression line.

```
ggplot(data = dataframe, aes(x = residuals_2, y = residuals_1)) +
  geom_point(size = 2, shape = 2) +
  geom_smooth(method = lm, color = "green") +
  labs(title = "Association between Residuals of Presidential and Incument Models",
       x = "residuals 2",
       y = "residuals 1") +
  theme_bw()
```



3. Write the prediction equation.

To make the prediction equation, we need to find the relevant coefficients.

```
regression4
```

We get the following coefficients:

Coefficients:

```
(Intercept)  residuals_2  
-1.942e-18    2.569e-01
```

We can now write the prediction equation $\gamma = \alpha + \beta \cdot x$.

Residuals2 = Intercept + slope * residuals1

Residuals2 = 0 + 0.256 * residuals1

On average, for every one-unit increase of "residuals1" the output variable "residuals2" increases by 0.3880.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
regression5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)  
summary(regression5)
```

This gives us the following coefficients:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.4486442	0.0063297	70.88	<0.00000000000000002	***
difflog	0.0355431	0.0009455	37.59	<0.00000000000000002	***
presvote	0.2568770	0.0117637	21.84	<0.00000000000000002	***

2. Write the prediction equation.

We can now write the prediction equation $\gamma = \alpha + \beta \cdot x$.

Voteshare = Intercept + β_1 * difflog + β_2 * presvote

Voteshare = 0.448 + 0.035 * difflog + 0.256 * presvote

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

The coefficients of "presvote" in regressions 4 and 5 are the same because both regressions answer the same question in two different ways: how much the presidents' voteshare influences voteshare for incumbent president if account for campaign spending.